QuakeFloat8 (QF8)

A Provably Optimal Log-Domain 8-Bit Format for ML Multiply-Accumulate

February 2026

Noah Everett

Abstract

QuakeFloat8 (QF8) is a block-scaled logarithmic 8-bit number format for machine learning in which multiplication reduces to 7-bit integer addition. The format uses a 1-bit sign, a 7-bit `u3.4` fixed-point $\log_2$ code per element, and a shared E8M0 block exponent per 32 elements. This document consolidates all theoretical results, hardware cost analysis, training experiments, Lean 4 verification results, and literature positioning for the QF8 project.

Key results: (1) Log-uniform quantization is provably minimax-optimal for NMSE over all source distributions (Theorem 3.5). (2) QF8 achieves +6.6 dB SQNR over IEEE FP8 E4M3 at the same e□ective storage cost. (3) The QF8 multiplier costs 66 gates vs. 231 for FP8 (3.5× cheaper). (4) TinyGPT-2 training with STE-based QAT matches FP32 validation loss. All theorems have been corrected, numerically verified, and checked in Lean 4.

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BMD Realizability

Definition 2.2 (Bit-Manipulation Decodable (BMD)). A decoder $\hat{D} : \{0,1\}^B \to R$ is bit-manipulation decodable if it can be expressed as a fixed-length straight-line program over $\{+, -, \times, \quad , \quad , \&, |, \quad \}$ on machine-word integers, with the output reinterpreted as a float via type punning.

Theorem 2.3 (BMD Realizability). The log-uniform decode function $\hat{D}(n) = a \cdot r^n$ can be implemented via:

1. Lookup table: $O(N)$ storage, $O(1)$ read.
2. Reinterpret cast (Quake trick): $I = \alpha n + \beta$, then oat reinterpret. Cost: 1 multiply + 1 add + type pun = 3 ops.
3. Quadratic approximation: $2^{e+f} = 2^e \cdot (1 + 0.6602f + 0.3398f^2)$. Cost: 2 multiplies + 2 adds + 1 shift = 5 ops. Maximum relative error: 0.27% (minimax optimal for constrained quadratic).

Theoretical Results: Minimax Optimality

Notation and Setup

Let X be a positive random variable with pdf f supported on $[a, b] \quad R_{>0}$. A B-bit scalar quantizer is a map $Q : [a, b] \to \hat{X}$ with $N = 2^B$ reconstruction points partitioning $[a, b]$ into cells $\{S_k\}_{k=0}^{N\mathbb{I}}$ with codepoints $\{c_k\}$.

Definition 3.1 (Distortion Measures).

$$\delta_X = X - Q(X), \qquad \varepsilon_X = \delta_X / X \tag{3}$$

$$\mathrm{MSE}_X = E[\delta_X^2], \qquad \mathrm{NMSE}_X = \frac{E[\delta_X^2]}{E[X^2]} \tag{4}$$

$$\mathrm{MSRE}_X = E[\varepsilon_X^2], \qquad \mathrm{SQNR} = 10 \log_{10}(1/\mathrm{NMSE}) \text{ dB} \tag{5}$$

Definition 3.2 (Auxiliary Quantities).

$$\alpha_X = E[X\delta_X]/E[X^2] \qquad \text{(signal-error correlation)} \tag{6}$$

$$\beta_X = E[Q(X)\delta_X]/E[X^2] \qquad \text{(reconstruction-error correlation)} \tag{7}$$

$$\gamma_X = E[Q(X)^2]/E[X^2] \qquad \text{(quantized power ratio)} \tag{8}$$

$$\rho_X = E[X \cdot Q(X)]/E[X^2] \qquad \text{(signal-quantized correlation)} \tag{9}$$

Algebraic identities (no assumptions required):

$$\alpha_X = \beta_X + \mathrm{NMSE}_X, \qquad \gamma_X = 1 - 2\alpha_X + \mathrm{NMSE}_X, \qquad \rho_X = 1 - \alpha_X \tag{10}$$

Definition 3.3 (Centroid Condition). A quantizer satisfies the centroid condition if $E[\delta_X \mid X \quad S_k] = 0$ for every cell. This holds for Lloyd-Max quantizers and approximately for any well-designed quantizer in the high-resolution regime. Under the centroid condition: $\beta_X = 0$, $\alpha_X = \mathrm{NMSE}_X$, and $\gamma_X = 1 - \mathrm{NMSE}_X$.

Definition 3.4 (Log-Uniform Quantizer). The log-uniform quantizer with N levels over dynamic range $[a, b]$ has cell boundaries $a \cdot r^k$ for $k = 0, \ldots, N$, where $r = (b/a)^{1=N} = 2^{R=N}$ and $R = \log_2(b/a)$ is the dynamic range in octaves. With geometric midpoint codepoints $c_k = a \cdot r^{k+1=2}$, the relative cell width is constant:

$$\frac{w(x)}{x} = r - 1 = e^{\shortparallel} - 1 \approx \varepsilon, \qquad \varepsilon = \frac{R \ln 2}{N} \tag{11}$$

where $\alpha_X = E[X\delta_X]/E[X^2]$. Equivalently:

$$\mathrm{NMSE}_{\mathrm{prod}} = 1 - 2\rho_X\rho_Y + \gamma_X\gamma_Y \tag{15}$$

This is exact, using only $X \perp Y$. No high-resolution, centroid, or distributional assumptions.

Proof. **Step 1.** Decompose the product error:

$$XY - Q_X Q_Y = (Q_X + \delta_X)(Q_Y + \delta_Y) - Q_X Q_Y = \underbrace{Q_X\delta_Y}_{A} + \underbrace{\delta_X Q_Y}_{B} + \underbrace{\delta_X\delta_Y}_{C}$$

Step 2. Expand $E[(A+B+C)^2]$ using $X \perp Y$. Since Q_X, δ_X depend only on X and Q_Y, δ_Y depend only on Y, every product of an X-function and a Y-function factors:

$$\begin{aligned} E[(A+B+C)^2] &= E[Q_X^2]E[\delta_Y^2] + E[Q_Y^2]E[\delta_X^2] + E[\delta_X^2]E[\delta_Y^2] \\ &\quad + 2E[Q_X\delta_X]E[Q_Y\delta_Y] + 2E[Q_X\delta_X]E[\delta_Y^2] + 2E[\delta_X^2]E[Q_Y\delta_Y] \end{aligned}$$

Step 3. Divide by $E[X^2]E[Y^2]$ and substitute $\beta_X = \alpha_X - \mathrm{NMSE}_X$, $\gamma_X = 1 - 2\alpha_X + \mathrm{NMSE}_X$:

$$\mathrm{NMSE}_{\mathrm{prod}} = \gamma_X\mathrm{NMSE}_Y + \gamma_Y\mathrm{NMSE}_X + \mathrm{NMSE}_X\mathrm{NMSE}_Y + 2\beta_X\beta_Y + 2\beta_X\mathrm{NMSE}_Y + 2\mathrm{NMSE}_X\beta_Y$$

Expanding $\beta = \alpha - \mathrm{NMSE}$ and simplifying yields (14). □ □

Centroid Simplification

Corollary 4.2 (Product Error Under Centroid Condition). If both quantizers satisfy the centroid condition ($\alpha_X = \mathrm{NMSE}_X$, $\alpha_Y = \mathrm{NMSE}_Y$), then:

$$\boxed{1 - \mathrm{NMSE}_{\mathrm{prod}} = (1 - \mathrm{NMSE}_X)(1 - \mathrm{NMSE}_Y)} \tag{16}$$

Equivalently: $\mathrm{NMSE}_{\mathrm{prod}} = \mathrm{NMSE}_X + \mathrm{NMSE}_Y - \mathrm{NMSE}_X \cdot \mathrm{NMSE}_Y$.

Proof. Substitute $\alpha_X = n_X$, $\alpha_Y = n_Y$ into (14): $n_X + n_Y + n_X n_Y + 2n_X n_Y - 2n_X n_Y - 2n_X n_Y = n_X + n_Y - n_X n_Y$. □ □

Remark (Sign of the Cross-Term). The correct sign is **minus** ($n_X + n_Y - n_X n_Y$), not plus as in the original formulation. The identity $(1-n_X)(1-n_Y) = 1 - n_{\mathrm{prod}}$ has a clean interpretation: the signal preservation fraction of the product equals the product of individual preservation fractions. However, since $n_X n_Y = O(1/N^4)$ while $n_X + n_Y = O(1/N^2)$, the distinction is negligible in practice.

Remark (Correction from Original). The original Theorem 3.1 used an MSRE-based formula with an invalid factorization ($E[X^2\varepsilon_X^2] = E[X^2]E[\varepsilon_X^2]$, which fails because ε_X depends deterministically on X). For non-log-uniform quantizers, the MSRE formula can be off by factors of 6–1000×. The corrected formula (14) uses only the valid factorization $X \perp Y$.

Specialization to Log-Uniform Quantizers

Theorem 4.3 (Log-Uniform Product Error). For log-uniform quantizers with geometric midpoints, MSRE = NMSE = $\varepsilon^2/12$ for any density, and:

$$\mathrm{NMSE}_{\mathrm{prod}} = \frac{(R\ln 2)^2}{6N^2} + O(1/N^4) = 2\mathrm{NMSE}^{\Lambda} + O(1/N^4) \tag{17}$$

The original MSRE-based formula gives the correct answer to leading order for log-uniform quantizers (despite being invalid in general) because the equalization property makes MSRE ≈ NMSE with ratio 1.0009 for $N = 16$.